

H-60 Phase 2 in Plain English

What this is: the non-technical summary of Phase 2 of the H-60 improvement plan, "delete first, refactor second". Written 8 October 2026, while the phase is in progress. The technical record is `docs/phase-2/PHASE_2_PLAN.md`.

Where we are

Phase 0 fixed the real bugs and shipped as Stable 0.7.9 in September. Phase 1 untangled who owns the aircraft's configuration and shipped as Stable 0.7.10 on 6 October. Then something arrived ahead of schedule: Brad's **HeliSim** flight model. It replaces the aircraft's engines, electrics, fuel and flight physics with a proper simulation, it is on the Dev Branch now as build 0.8, and it will become the next Stable release.

Phase 2 is being done on top of HeliSim rather than waiting for it. That decision was made on 7 October. HeliSim removes three whole parts of the mod and edits two dozen more, so cleaning up the old tree first would have meant cleaning up things HeliSim was already throwing away.

What Phase 2 is for

The mod has grown over several years and several authors. Along the way it collected files that nothing uses, references with the wrong capitalisation that only work because Windows does not care, duplicate copies of the same texture, and thousands of lines of code that were commented out and never came back. None of it hurts players today. All of it gets in the way of the next step, which is moving the mod to a modern build system that runs on Linux and checks our work automatically. Linux does care about capitalisation, and a build checker cannot tell dead code from live code.

So Phase 2 removes what nothing uses and fixes what only Windows tolerates. The rule for the whole phase is **nothing changes for the player**. Every change is proven by building the mod before and after and comparing the result byte for byte.

What has been done

Four changes are finished and waiting to be reviewed and merged. Each is small on purpose, so a reviewer can read it in one sitting.

1. Fix the spelling of file references. Eleven places in the mod referred to a file with the wrong capitalisation. We corrected the references to match the files. We also found two copies of the same door-gun definition and kept one.

2. Remove dead files. Thirty-seven files that nothing in the mod ever reads: empty placeholder files, old versions of cockpit display pages that were replaced years ago but never deleted, a leftover copy of a crew animation, a duplicate map texture, and two pieces of build configuration for tools we no longer use. About 2.6 MB.

3. Clear out commented-out code. Around 440 lines of code that had been switched off by turning it into comments, in 25 files, plus two files that consisted of nothing but a commented-out block. Where a comment documented an experiment that failed, we kept it and said so, because that is useful to the next person. We also corrected three comments that gave a wrong reason for a design choice.

4. Share duplicate textures. Three livery textures were exact copies of another texture in the mod. The liveries now point at the one copy, which takes 10.5 MB out of the download with no visible change.

While checking the liveries in game we found an actual bug and fixed it in the same change: applying any livery in the Virtual Garage painted white "UNITED STATES ARMY" lettering on the tail and a white number on the engine cover. The cause was a blank entry in every livery definition. The UH-60M and the Slick now spawn with the proper US Army skin and dark lettering, and the broken "UH-60M" entry in the livery list is gone.

How we know nothing changed

For each change, the affected parts of the mod were built twice, once before and once after, and the compiled configuration files were compared. They came out identical apart from a timestamp the build tool writes. The one change that could not be byte-identical, the texture sharing, was checked in game by Riverman: every affected livery, and a full walk through every cockpit display page, since display files were among the deletions.

What is left in this phase

- **Three copies of the display definitions** that should be one. Brad is editing one of them for HeliSim's cockpit, so merging them now would fight his work. This waits until his edits settle.
- **Eleven display elements switched off with a condition.** Removing them changes the compiled files, so they need the in-game check and a decision on whether any was a half-built feature.
- **Display definitions that only the removed MLASS variant used.** They go once the MLASS retirement reaches the main branch.
- **A sweep of the model's own folder:** some crew animations nothing references, two duplicated cockpit textures, and a dozen copies of one reflection map. Each needs a check that the 3D model itself does not use it.
- **A commented-out block of radio controls** that looks parked rather than abandoned. We will ask BroBeans before touching it.
- **The long tail** of files with one or two commented lines each. Low value, and not worth the review noise unless someone wants the tree spotless.

What it means for players

Nothing you can see, except the livery fix and a slightly smaller download. The point of Phase 2 is what comes after it: a build that can run on a server, check itself, and refuse mistakes, which is Phase 3.